TEXORA · COURSE NOTES · LECTURE SHEET

Matrices

Arrays of numbers and the algebra that works on them — types, operations, determinants, inverses, rank, and the solution of linear systems.

MATH-1101 · MATHEMATICS-I · TOPIC 1 · LEVEL 1 TERM 1

1. What a matrix is

A **matrix** is a rectangular array of numbers (or symbols) arranged in rows and columns, enclosed in brackets. A matrix with *m* rows and *n* columns is of **order m × n**, and its entries are written a_{ij}, where *i* names the row and *j* the column. Matrices are the working language of every subject that handles many quantities at once — systems of equations in engineering, stress and strain in mechanics, and data tables everywhere. Two matrices are **equal** only when they have the same order and every corresponding entry is equal.

1.1 The standard types

TYPE	DEFINITION
Row / column matrix	A single row (1 × n) or a single column (m × 1).
Square matrix	m = n; the entries a_{11}, a_{22}, … form the *principal diagonal*.
Diagonal matrix	Square, with every off-diagonal entry zero.
Scalar / identity matrix	Diagonal with all diagonal entries equal (scalar), or all equal to 1 (identity, written I).
Zero (null) matrix	Every entry zero, of any order.
Upper / lower triangular	Square, with all entries below (upper) or above (lower) the diagonal equal to zero.
Symmetric	$a_{ij} = a_{ji}$ for all i, j — the matrix equals its transpose.
Skew-symmetric	$a_{ij} = -a_{ji}$; the diagonal entries are necessarily zero.
Orthogonal	A square matrix with $A\,A^{T} = I$, so its inverse equals its transpose.

2. Operations on matrices

- **Addition and subtraction** — only between matrices of the same order, entry by entry. Addition is commutative and associative.
- **Scalar multiplication** — every entry multiplied by the scalar.
- **Matrix multiplication** — the product AB exists only when the number of columns of A equals the number of rows of B. The entry $(AB)_{ij}$ is the dot product of row *i* of A with column *j* of B. Multiplication is associative and distributive, but **not commutative**: in general $AB \neq BA$, and $AB = 0$ does not imply $A = 0$ or $B = 0$.
- **Transpose** A^T — rows and columns exchanged. Key rules: $(A + B)^T = A^T + B^T$, $(AB)^T = B^T A^T$ (note the reversed order), $(kA)^T = kA^T$.

WORKED EXAMPLE — MULTIPLICATION

Let $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 5 \\ 0 & 2 \end{pmatrix}$. Then AB: entry (1,1) = 2·1 + 1·0 = 2; (1,2) = 2·5 + 1·2 = 12; (2,1) = 3·1 + 4·0 = 3; (2,2) = 3·5 + 4·2 = 23. So $AB = \begin{pmatrix} 2 & 12 \\ 3 & 23 \end{pmatrix}$. (BA would give a different matrix — check it as an exercise.)

3. Determinants

Every square matrix A has a number attached to it, its **determinant** |A| or det A. For 2 × 2, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has determinant $ad - bc$. For 3 × 3 and larger, the determinant is found by **cofactor expansion** along any row or column: each entry is multiplied by its cofactor — the determinant of the smaller matrix left after deleting its row and column, with the chessboard sign $(-1)^{i+j}$ — and the products are added.

WORKED EXAMPLE — A 3 × 3 DETERMINANT

Expand along the first row: | 1 2 3 ; 0 1 4 ; 5 6 0 | = 1·(1·0 − 4·6) − 2·(0·0 − 4·5) + 3·(0·6 − 1·5) = 1·(−24) − 2·(−20) + 3·(−5) = −24 + 40 − 15 = **1**.

Useful properties. Interchanging two rows changes the sign; a row of zeros, or two equal or proportional rows, makes the determinant zero; multiplying a row by k multiplies the determinant by k, so $\det(kA) = k^n \det A$ for order n; $\det(AB) = \det A \cdot \det B$; $\det(A^T) = \det A$; adding a multiple of one row to another leaves the determinant unchanged — the property that makes row reduction a practical way to evaluate large determinants.

4. Inverse of a matrix

A square matrix A is **non-singular** when $\det A \neq 0$, and then it possesses a unique **inverse** A^{-1} with $A\,A^{-1} = A^{-1}\,A = I$. If $\det A = 0$ the matrix is **singular** and no inverse exists. The inverse is built from the **adjugate** (the transpose of the cofactor matrix):

$$A^{-1} = (1 / \det A) \cdot \operatorname{adj} A$$

WORKED EXAMPLE — INVERTING A 2 × 2 MATRIX

$A = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$; $\det A = 24 - 14 = 10$. For 2 × 2, swap the diagonal entries and change the signs of the off-diagonal ones: $\operatorname{adj} A = \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$, so $A^{-1} = (1/10)\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$. Multiplying A by A^{-1} returns I, as it must.

5. Rank

The **rank** of a matrix is the order of the largest non-vanishing minor (sub-determinant), equivalently the number of independent (non-zero) rows after the matrix is reduced to row-echelon form by elementary operations. Rank measures how much independent information the matrix carries, and it decides the consistency of linear systems: the system AX = B has a solution exactly when rank(A) = rank(augmented matrix [A | B]), and the solution is unique when this common rank equals the number of unknowns.

6. Systems of linear equations

Three classical methods solve A X = B. **Matrix inversion** gives $X = A^{-1}B$ directly when A is non-singular. **Cramer's rule** writes each unknown as a ratio of determinants: $x_j = \det A_j / \det A$, where A_j is A with its j-th column replaced by B — elegant for small systems. **Gaussian elimination** reduces the augmented matrix to echelon form and back-substitutes; it is the method that scales to large systems and the one computers use.

WORKED EXAMPLE — CRAMER'S RULE

Solve $2x + 3y = 8$, $x - y = 1$. $D = (2)(-1) - (3)(1) = -5$. $D_x = (8)(-1) - (3)(1) = -11$, so $x = (-11)/(-5) = 2.2$. $D_y = (2)(1) - (8)(1) = -6$, so $y = (-6)/(-5) = 1.2$. Check: $2(2.2) + 3(1.2) = 8$ ✓ and $2.2 - 1.2 = 1$ ✓.

7. Eigenvalues and eigenvectors (introduction)

A non-zero vector X is an **eigenvector** of a square matrix A when A X = λ X for some number λ: the matrix stretches the vector without turning it. The λ values are the **characteristic roots (eigenvalues)**, found from the characteristic equation det(A − λI) = 0. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, det(A − λI) = (2 − λ)² − 1 = 0 gives λ = 1 and λ = 3. Eigenvalues govern vibration, stability and stress problems throughout engineering, and return in later courses.

8. Gaussian elimination in full — one complete reduction

Because elimination is the method that survives contact with large systems, here is one complete run. Solve x + 2y + z = 7, 2x − y + z = 3, x + y − z = 2. The augmented matrix is [1 2 1 | 7; 2 −1 1 | 3; 1 1 −1 | 2]. Subtract 2×row 1 from row 2 and row 1 from row 3: [1 2 1 | 7; 0 −5 −1 | −11; 0 −1 −2 | −5]. Interchange rows 2 and 3 for convenience: [1 2 1 | 7; 0 −1 −2 | −5; 0 −5 −1 | −11]. Subtract 5×row 2 from row 3: the third row becomes (0, 0, −1 + 10, −11 + 25) = (0, 0, 9, 14), so z = 14/9. Back-substitute: −y − 2z = −5 gives y = 5 − 2z = 5 − 28/9 = 17/9. Then x = 7 − 2y − z = 7 − 34/9 − 14/9 = 7 − 48/9 = 15/9 = 5/3. Verification in the second equation: 2(5/3) − 17/9 + 14/9 = 30/9 − 17/9 + 14/9 = 27/9 = 3 ✓. The pattern to absorb: reduce to a triangle with row operations that never change the solution set, then read the answer off backwards.

9. Key points and summary

- A matrix is an ordered array; operations respect order and shape — addition needs equal orders, multiplication needs matching inner dimensions, and AB ≠ BA in general.
- The determinant compresses a square matrix into one number: it tests singularity, scales with row operations by known rules, and multiplies over products.
- The inverse A^{-1} = adj A / det A exists exactly when det A ≠ 0, and solves systems by $X = A^{-1}B$. Rank decides consistency and uniqueness in general.
- Cramer’s rule, elimination and inversion are three views of the same solution; eigenvalues open the door to the matrix methods of later engineering courses.

10. Practice

Work each problem fully; final answers are given at the end.

1. For A = [1 2; 3 4] and B = [2 0; 1 3], compute AB and BA, and confirm they differ.
2. Evaluate the determinant of [2 5; 1 3].

3. Find the inverse of [3 1; 2 1].
4. Solve by any method: x + y = 5, 2x − y = 4.
5. State the rank of [1 2; 2 4] and justify it in one line.
6. Find the eigenvalues of [4 1; 2 3].

Answers

1. AB = [4 6; 10 12]; BA = [2 4; 10 14] — different, as expected.
2. 6 − 5 = 1.
3. det = 1, so the inverse is [1 −1; −2 3].
4. x = 3, y = 2.
5. Rank 1 — the second row is twice the first, so only one row is independent (equivalently, the 2 × 2 determinant is 0 and a 1 × 1 minor is not).
6. $\lambda^2 - 7\lambda + 10 = 0$, so $\lambda = 2$ and $\lambda = 5$.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib